Nathan S. Ryan

Contact Information

Nuclear Fuel Cycle and Advanced Reactor Analyst
Nuclear Innovation Alliance

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Research Interests

Nuclear fuel cycle simulation and analysis, geographical information systems, spatial modeling, energy regulation, supply chain analysis, nuclear energy policy, hydrogen fuel, and energy system modeling.

Experience

Nuclear Innovation Alliance, Washington, DC
Analyst, **September 2026 – Present**

Oak Ridge National Laboratory, Oak Ridge, TN
Research Scientist, **June 2025 – August 2026**
Graduate Summer Research Assistant, **June 2024 – Aug 2024**
Reactor and Nuclear Fuel Cycle Analysis Group

University of Illinois at Urbana-Champaign, Urbana, IL
Research Assistant, Nuclear Plasma and Radiological Engineering **Aug 2022 – May 2025**
Undergraduate Research Assistant **May 2020 – Aug 2022**
Advanced Reactors and Fuel Cycles group.

MS

University of Illinois Urbana-Champaign, **Aug 2022 – May 2025**
Nuclear, Plasma, and Radiological Engineering

BS

University of Illinois Urbana-Champaign, **Aug 2019 – Aug 2022**
Engineering Physics

Honors and Awards

NPRE, Students Award for Excellence in Graduate Service	**2023**
ANS, Student Sections Committee Commendation for Service and Leadership	**2023**
NPRE, Students Award for Excellence in Undergraduate Service	**2022**
UIUC, Dean's List	**2020**
Department of Physics, Excellence in Physics Award	**2020**

Journal Publications

[1] A. M. Bachmann, R. Fairhurst-Agosta, Z. Richter, **N. S. Ryan**, and M. Munk. Enrichment dynamics for advanced reactor HALEU support. *EPJ Nuclear Sciences and Technologies*, 7:22, 2021. Publisher: EDP Sciences. URL: https://www.epj-n.org/articles/epjn/abs/2021/01/epjn210024/epjn210024.html, doi:10.1051/epjn/2021021

Refereed Conference Abstracts

[2] Y. Robert, J. W. Bae, E. E. Davidson, **N. S. Ryan**, A. M. Bachmann, T. K. Kim, S. Richards, and J. Zhou. Quantifying cost drivers and uncertainties in nuclear fuel cycle scenarios through integrated modeling. In *ANS Global 2026*, 2026

[3] **N. S. Ryan**, K. D. Huff, and M. Munk. Leu+ to haleu transitions in advanced reactor fuel cycles. In A. M. Bachmann, editor, *Technical Workshop on Fuel Cycle Simulation 2025 (TWFCS 2025)*, 2025

Technical Reports

[4] S. G. Dotson, A. M. Bachmann, Z. M. Richter, N. R. Panczyk, **N. S. Ryan**, A. C. Balla, and E. R. Fanning. Economic and Carbon Impacts of Potential Illinois Nuclear Plant Closures: The Cost of Closures. Technical Report UIUC-ARFC-2021-02, University of Illinois at Urbana-Champaign, Urbana, IL, May 2021. URL: github.com/arfc/2021-04-nm-illinois

Poster Presentations

[5] **N. S. Ryan** and J. W. Bae. Enrichment and core-loading versatility in cyclus reactors. Poster presented at ORNL Summer Symposium, 2024. Oak Ridge, TN

[6] **N. S. Ryan** and N. Panczyk. Using an energy model to demonstrate the potential impact of proactive nuclear activism. Poster presented at *American Nuclear Society Student Conference*, 2022. Champaign, IL

Other Publications

[7] **N. S. Ryan**. LEU+ to HALEU nuclear fuel cycle transitions and dynamic reactor models. Master's thesis, University of Illinois at Urbana-Champaign, May 2025. URL: https://hdl.handle.net/2142/129324

Opinion Pieces

[8] K. D. Huff and **N. S. Ryan**. What it will take to restart decommissioned US nuclear plants. A primer, Oct. 2024. URL: https://thebulletin.org/2024/10/what-it-will-take-to-restart-decommissioned-us-nuclear-plants-a-primer/

Invited Talks

ORNL Research Symposium, Updates to time, power, and deployment in, advanced reactor fuel cycle modeling, *Speaker*. **March 27, 2025**
ANS U. Illinois, Presenting at a Conference, *Speaker*. **March 10, 2025**
ANS GL-Chicago, Cyclus, LEU+, and HALEU fuel cycles, *Presenter*. **January 30, 2025**
ANS U. Illinois, Perfecting a Conference Abstract, *Speaker*. **Dec 9, 2024**
ANS U. Illinois, How to Present at a Conference, *Speaker*. **March 13, 2023**
ANS U. Illinois, How to Create a Conference Abstract, *Speaker*. **Jan 23, 2023**
U. Illinois, Physics and Astronomy Convocation, *Speaker*. **May 15, 2022**
Tinley Park High School, Introduction to Nuclear Science & Careers, *Speaker*. **Jan 17, 2020**
TEDxYouth, Evolutionary Anthropology and Youth in Science, *Speaker*. **Feb 15, 2019**

Panels & Workshops

ANS Midwest, Attending your First Conference, *Panelist*. **November 9, 2024**
ANS Midwest, Nuclear Outreach with Social Media, *Workshop Lead*. **November 9, 2024**
Institute for Genomic Biology, Boundaries in Graduate School, *Panelist*. **Aug 24, 2023**
ANS, Social Media for Local and Student Section Members Webinar, *Panelist*. **Aug 3, 2023**
Memphis Rise Academy, Careers in Nuclear, *Panelist*. **May 31, 2023**
U. Tennessee, Knoxville, Navigating Conversations with Advisors, Managers, and Supervisors, *Moderator*. **April 14, 2023**

Scientific Computing Teaching

Computational Resource Access Network, Board of Directors **2022 – Present**
Introduction to Python, Numerical Methods, Parallel Programming, and Best Practices

Worldwide Youth in Science and Engineering, Urbana, IL **July 2022, 2023**
Introduction to Python, and Best Practices for Beginners

Scientific Computing Skills

Coding Languages Bash, C++, Python, HTML, CSS, LaTeX, BibTeX, XML
Build Systems make, CMake, Docker
Databases SQL, SQLite, JSON, CSV
Test Frameworks PyTest
Version Control git
Applications Sphinx, Mathematica, PyTorch, Word, Excel, Powerpoint, cPanel
Nuclear Codes Cyclus, Serpent, SCALE

Professional Service

Mentor, Appalachian STEM Academy, ORNL **2026**
Member, Student Sections Committee, ANS **2023 – 2026**
Delegate, Nuclear Engineering Student Delegation **2023, 2024**
Vice Chair, Student Sections Committee, ANS **2023 – 2025**
Subcommittee Chair, Student and Young Member Liaison Committee, ANS **2023 – 2025**
Subcommittee Chair, Student Conference Committee, ANS **2022 – 2025**
Chair, Winter Meeting Student Program, ANS **2022**
Organizer, Young Professionals Congress, ANS **2022 – 2023**
Member, ANS CARES, ANS **2022 – 2026**
Public Relations Officer, U. Illinois ANS Student Section **2020 – 2022**
Coordinator, Engineering Outreach Bureau, Grainger Engineering Council **2020 – 2022**
Media Chair, 2022 ANS Student Conference **2019 – 2022**

Departmental Service	**Committee Member**, Diversity, Equity, and Inclusion Committee	**2022 – 2025**
	Committee Chair, Graduate Student Advisory Committee	**2022 – 2023**
	Camp Advisor, Worldwide Youth in Science and Engineering	**2022 – 2024**

References *Available upon request*